

MID-WAY, TX, USA

WMO: 720299

Lat: 32.456N

Lon: 96.913W

Elev: 749

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.4	(c) 27.9	(d) 11.5	(e) 10.2	(f) 33.2	(g) 15.7	(h) 12.5	(i) 36.9	(j) 26.7	(k) 54.0	(l) 24.7	(m) 52.6	(n) 7.2	(o) 350	(p) 0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 21.8	(c) 100.3	(d) 72.7	(e) 98.8	(f) 72.6	(g) 96.4	(h) 73.0	(i) 77.9	(j) 89.4	(k) 77.2	(l) 88.5	(m) 76.5	(n) 87.5	(o) 7.8	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	136.9	79.9	74.6	133.0	79.8	73.5	128.3	79.7	41.7	89.1	41.0	88.6	40.3	87.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.0	(b) 19.9	(c) 17.8	(e) 16.8	(f) 103.4	(g) 4.4	(h) 2.6	(i) 13.6	(j) 105.2	(k) 11.0	(l) 106.7	(m) 8.5	(n) 108.2	(o) 5.3	(p) 110.1		
(a) 23.0	(b) 19.9	(c) 17.8	(e) 16.2	(f) 78.6	(g) 4.7	(h) 1.9	(i) 12.9	(j) 80.0	(k) 10.1	(l) 81.1	(m) 7.5	(n) 82.2	(o) 4.1	(p) 83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	46.6	49.8	58.2	64.8	72.7	80.9	83.9	84.6	78.4	67.1	56.4	47.9
	DBStd	15.58	9.89	10.42	9.24	7.40	5.97	3.87	3.74	4.55	5.64	8.52	9.33	9.21
	HDD50	541	181	123	35	4	0	0	0	0	0	3	43	152
	HDD65	2247	571	433	244	92	13	0	0	0	0	79	282	533
	CDD50	6395	77	118	289	449	703	927	1052	1073	852	533	235	87
	CDD65	2626	2	8	33	87	251	477	587	608	402	143	24	4
	CDH74	28899	19	54	207	626	2100	5240	7236	7732	4233	1286	144	22
	CDH80	14383	0	4	24	126	724	2598	3960	4402	2074	444	24	3
(14) Wind	WSAvg	7.5	8.5	9.0	9.4	9.0	7.7	7.2	6.4	5.9	5.6	6.7	7.6	7.6
(15) Precipitation	PrecAvg	36.1	1.8	2.7	3.0	3.7	4.9	3.2	2.0	2.0	3.9	3.9	2.6	2.7
	PrecMax	59.0	4.8	5.8	7.3	11.3	10.1	8.2	6.7	5.9	9.7	11.1	6.3	8.5
	PrecMin	19.6	0.0	0.2	0.3	0.2	1.1	0.2	0.1	0.1	0.8	0.3	0.4	0.4
	PrecStd	7.8	1.4	1.5	1.9	2.8	2.8	2.1	1.9	1.7	2.6	3.3	1.5	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	79.8	82.3	88.3	93.3	99.7	103.5	104.3	99.6	92.5	83.2	76.9
		MCWB	59.1	61.6	64.6	68.2	73.1	72.0	73.1	71.4	70.4	71.4	66.1	62.3
	2%	DB	71.8	75.0	79.3	82.9	90.0	96.7	99.9	101.3	96.6	88.8	78.2	71.8
		MCWB	58.3	62.5	63.8	67.5	71.3	72.1	72.7	73.0	71.6	70.9	64.5	61.8
	5%	DB	67.6	70.4	75.9	80.7	87.1	94.2	97.6	99.3	93.4	85.0	73.8	67.7
		MCWB	57.7	60.2	62.8	66.3	71.5	73.3	73.1	73.0	72.0	69.3	63.6	60.4
	10%	DB	63.3	67.3	72.5	77.5	84.2	91.4	95.2	96.9	90.8	81.5	70.9	63.5
		MCWB	55.5	59.2	61.8	65.4	70.5	73.4	74.0	73.3	71.5	66.3	62.5	56.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.2	68.6	69.3	73.4	77.1	79.1	79.1	78.4	77.5	75.7	71.9	68.6
		MCDB	69.6	73.4	76.3	79.8	85.5	90.3	91.3	90.7	88.0	83.9	77.0	71.2
	2%	WB	63.0	66.1	67.5	71.2	75.3	77.7	77.9	77.4	76.6	74.2	68.8	65.3
		MCDB	66.7	70.3	74.3	78.5	83.1	87.9	89.9	89.8	86.6	82.2	72.7	69.2
	5%	WB	60.3	63.4	66.1	69.6	74.0	76.7	77.1	76.7	75.5	72.7	66.6	61.6
		MCDB	65.4	68.3	72.5	75.8	82.0	86.6	88.8	89.1	84.8	80.3	71.5	65.2
	10%	WB	55.9	60.0	64.3	68.1	72.7	75.6	76.2	76.0	74.5	70.9	64.2	58.3
		MCDB	62.4	65.7	70.1	74.9	80.6	85.3	87.6	88.3	83.6	78.1	69.5	62.8
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	19.7	20.5	20.9	19.3	20.1	21.1	21.8	21.7	21.9	20.3	19.2
		MCDBR	25.8	24.7	23.1	22.4	22.3	23.5	23.9	24.7	24.3	24.0	22.7	24.9
		MCWBR	15.6	15.0	12.3	11.2	8.0	5.9	5.4	5.5	6.9	9.5	12.7	16.1
	5% WB	MCDBR	21.4	21.0	18.8	18.1	18.0	18.4	20.1	20.8	19.7	20.1	18.1	20.0
		MCWBR	16.1	15.2	12.0	11.0	8.2	6.4	5.8	5.4	6.7	9.5	13.4	16.8
(40) Clear-Sky Solar Irradiance	taub		0.313	0.329	0.354	0.394	0.422	0.433	0.437	0.437	0.398	0.357	0.337	0.318
	taud		2.506	2.467	2.414	2.324	2.303	2.304	2.332	2.314	2.375	2.476	2.473	2.495
	Ebn at Noon		289	294	293	285	276	271	269	268	275	281	277	280
	Edh at Noon		27	31	36	41	42	42	40	41	36	31	28	26
(44) All-Sky Solar Radiation	RadAvg		918	1065	1413	1717	1850	2080	2126	1952	1643	1293	971	810
	RadStd		89	172	114	131	163	191	147	128	135	192	123	106

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page